

Reporting

Report Date: May 7th, 2025
Receipt Date: Apr 24th, 2025
Collection Date: Apr 23rd, 2025
Specimen: Blood
Status: FINAL

Krishna Alluri

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Additional Recipient: N/A



Complete Tumor Response Map on page 3

Summary of Detected Somatic Alterations & Biomarkers with Associated Treatment Options

		 Approved in indication	 Approved in other indication	 Lack of response	
DETECTED ALTERATION(S) / BIOMARKER(S)	ASSOCIATED FDA-APPROVED THERAPIES				CLINICAL TRIALS (SEE PAGE 4) % CFDNA OR COPY NUMBER
BRCA2 F2560fs	 Niraparib, Olaparib, Rucaparib, Talazoparib				Yes 50.5%
BCOR L1632fs	None				No 0.2%

Variants of uncertain clinical significance listed on following pages.

Additional Biomarkers

Tumor Mutational Burden (TMB)	Not Evaluable
MSI-High	NOT DETECTED
Tumor Fraction [#]	<0.05%
Suspected Germline Variant(s) [‡]	BRCA2 F2560fs The BRCA2 F2560fs (c.7679_7680delTT) alteration was detected in this patient's sample at an allele fraction suspicious for germline origin. This variant may lead to the loss of functional protein, and similar variants have been associated with hereditary predisposition to cancer.

[#] Tumor fraction is defined as the proportion of tumor molecules present in the cfDNA within the submitted specimen
[‡] As Guardant360 is neither intended nor validated for the reporting or interpretation of germline variants, we cannot confirm the germline vs. somatic origin of this finding and recommend verification with an assay validated for germline testing if this potential incidental finding is of clinical interest.

Comments

Reported by: VL

Variants of Uncertain Clinical Significance

CHEK1 S316N (0.1%)

FGFR1 R56_L57insCR (0.07%)

KDM4A K417R (0.2%)

NOTCH1 S2288S (0.2%)

Variants of Uncertain Clinical Significance: The functional consequences and/or clinical significance of these alterations is uncertain. Relevance of therapies targeting these alterations is unknown.

We evaluated this sample for MSI, TMB, and alterations in 740 genes, including the following guideline-recommended genes for NSCLC

ALK

BRAF

EGFR

ERBB2 (HER2)

KRAS

MET

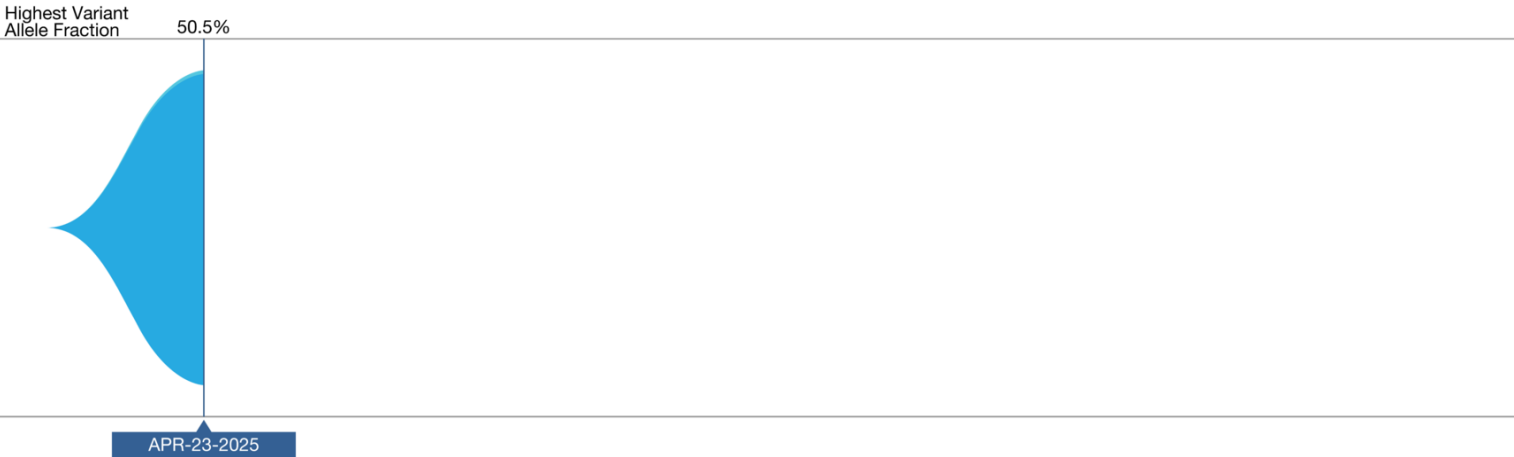
NTRK

RET

ROS1

Guardant360 Tumor Response Map

The Guardant360 Tumor Response Map illustrates the variant allele fraction (% cfDNA) of observed somatic variants at each sample submission. Amplifications are not plotted, and only the first and last five test dates are plotted. Please see the Physician Portal (portal.guardanthealth.com) for the Tumor Response Map with all test dates.



DETECTED ALTERATION(S) / BIOMARKER(S)	% CFDNA OR COPY NUMBER	
Tumor Fraction	<0.05%	
BRCA2 F2560fs	50.5%	
BCOR L1632fs	0.2%	
CHEK1 S316N	0.1%	Variants of Uncertain Clinical Significance §
FGFR1 R56_L57insCR	0.07%	Variants of Uncertain Clinical Significance §
KDM4A K417R	0.2%	Variants of Uncertain Clinical Significance §
NOTCH1 S2288S	0.2%	Variants of Uncertain Clinical Significance §

The table above annotates the variant allele fraction (% cfDNA) detected in this sample, listed in descending order.
§ See definitions section for more detail

Available Clinical Trials (within the same state as the ordering physician)

There may be additional trials not listed here. Visit: portal.guardanthealth.com or email clientservices@guardanthealth.com with A1376063 in the subject line of the email, for additional trials.

ALTERATION	TRIAL ID / CONTACT	TITLE	PHASE	SITE(S)
BRCA2 F2560fs	NCT02693535 Pam Mangat, MS, tapur@asco.org , www.tapur.org	TAPUR: Testing the Use of Food and Drug Administration (FDA) Approved Drugs That Target a Specific Abnormality in a Tumor Gene in People With Advanced Stage Cancer	Phase 2	Houston, TX
	NCT05327010 See https://clinicaltrials.gov/ct2/show/NCT05327010	Testing the Combination of the Anti-cancer Drugs ZEN003694 (ZEN-3694) and Talazoparib in Patients With Advanced Solid Tumors, The ComBET Trial	Phase 2	Houston, TX Austin, TX San Antonio, TX Galveston, TX
	NCT05700721 Timothy Yap, MBBS, PHD, tyap@mdanderson.org , (713) 563-1784	Phase II Trial of the PARP Inhibitor Niraparib and PD-1 Inhibitor Dostarlimab in Patients With Advanced Cancers With Active Progressing Brain Metastases (STARLET)	Phase 2	Houston, TX
	NCT05787587 IDEAYA Clinical Trials, IDEAYAClinicalTrials@ideayabio.com , 650-278-8351	A Study of PARG Inhibitor IDE161 in Participants With Advanced Solid Tumors	Phase 1	Houston, TX Irving, TX San Antonio, TX
	NCT05932862 Exelixis Clinical Trials, druginfo@exelixis.com , 1-888-393-5494	A Phase 1 Study of XL309 (ISM3091) Alone and in Combination in Patients With Advanced Solid Tumors	Phase 1	Houston, TX Austin, TX San Antonio, TX

More clinical trial options available at portal.guardanthealth.com

Interpretation

Somatic Alterations: Guardant360 detects somatic alterations in circulating cell-free DNA isolated from a patient's blood specimen in the genes listed in Table 1. These genomic alterations are cancer-associated somatic variants, some of which have been associated with either increased or reduced clinical response to specific treatments. The percentage of altered cell-free DNA circulating (% cfDNA) in blood is related to the unique tumor biology of each patient. Factors that may affect the % cfDNA of detected somatic alterations include tumor growth, turn over, size, heterogeneity, vascularization, disease progression, and treatment.

Somatic alterations may have additional annotations as follows:

Variants of Uncertain Clinical Significance: The functional consequences and/or clinical significance of alterations are unknown. Relevance of therapies targeting these alterations is uncertain.

Synonymous Alteration: This sequence change does not alter the amino acid at this position and is unlikely to be a therapeutic target. Clinical correlation is advised.

Clonal Hematopoiesis (CH): All SNVs and Indels detected on the Guardant 360 assay will be evaluated for the likelihood that they are derived from hematopoietic sources. The categorization of these variants as potential clonal hematopoietic origin by an algorithm is not intended to be used for the diagnosis of a hematolymphoid malignancy. The clinical significance of these variants requires evaluation of other clinical and laboratory parameters, such as peripheral blood counts, patient age, and treatment history. Non-tumor derived variants may arise from hematolymphoid tissues, such as those associated with clonal hematopoiesis, and/or include variants from clonal processes in other organ sites.

Nonsense Mutation: A point mutation that results in a premature stop gain.

Insertion (Ins): Guardant360 detects short insertions in exons of certain genes (see Table 1), including potential splice site-disrupting events.

Deletion (Del): Guardant360 detects short deletions in exons of certain genes (see Table 1), including potential splice site-disrupting events.

Indel: Guardant360 detects short insertions or deletions in the exons of certain genes (see Table 1). This type of variant represents frameshift, in-frame and complex alterations comprising both an insertion and a deletion. These variants can result in the substitution of one or more amino acids with the reading frame preserved (in-frame), or disrupt the reading frame (frameshift) leading to a premature stop codon.

Reversion: Reversions are secondary mutations, often small indels, that convert an initial frameshift mutation into an in-frame internal deletion that still produces a partly functional protein product.

MET exon 14 skipping: This type of genomic alteration is predicted to result in the skipping of *MET* exon 14 during RNA splicing, resulting in oncogenic activation of the *MET* protein.

Promoter Methylation: Guardant360 detects promoter methylation in the genes listed in Table 1. Hypermethylation of normally unmethylated CpG islands located in gene promoters correlates with transcriptional repression that can serve as an alternative to coding region mutations for inactivation of tumor suppressor genes (Baylin et al., 1997). DNA methylation of key CG dinucleotides within the promoter blocks accessibility of transcription factors to their binding motifs, thereby silencing gene expression. DNA hypermethylation occurs early in tumorigenesis and typically increases with tumor progression (Ehrlich 2019).

Splice Site: Splice site variants disrupt the donor and/or acceptor splice site(s), leading to abnormal mRNA splicing and altered protein levels and/or function.

Somatic Alterations Not Detected (ND): In some cases, Guardant360 does not detect any somatic alterations. Somatic alterations may be present that are below the limit of detection of this test. Certain sample or variant characteristics may result in reduced analytic sensitivity. The absence of detectable somatic alterations in circulating cell-free DNA does not preclude the presence of somatic alterations in the tumor.

Amplification: Guardant360 detects amplifications in the genes listed in Table 1. Gene amplification results in increased copies of the gene present in the cfDNA. The reported absolute copy number value represents the average copy number for the detected gene that was detected in circulating cfDNA. With the exception of sex-linked genes such as *AR*, 2 copies are expected in the absence of amplification. As the absolute number of copies in circulation is dependent on both tumor fraction and the magnitude of the tumor amplification, amplifications are reported on a semi-quantitative scale.

Amplifications are reported in two levels:

Medium (++): Amplification magnitude is below the 50th percentile of amplifications detected by Guardant360.

High (+++): Amplification magnitude is above the 50th percentile.

Unlike tissue based gene amplification tests (e.g. IHC or FISH), Guardant360 assesses the total representation of a given gene in all circulating cell-free DNA present in the patient's blood sample including material derived from the tumor and healthy tissue alike. As such, the absolute level of amplification present in the blood depends both on the tumor-derived cfDNA content and on the degree of amplification within that fraction and cannot be inferred from bulk cfDNA interrogation. For example, a positive Guardant360 test could represent a small population of cells with extremely high levels of the detected gene amplification. Alternatively, it could represent a large population of cells with low to medium levels of the detected gene amplifications. The exact correlation between amplification detected by Guardant360 compared to IHC or FISH and how each test differentially guides patient management is an area of active investigation.

Rearrangement: Guardant360 detects rearrangements in the genes listed in Table 1. Rearrangement refers to large genomic events that result in single or multi-exon deletions (>3Kb in size) or gene fusions.

Single Copy Deletion: Occurs when a single copy of a gene is lost due to a whole gene deletion.

Single Copy Deletion + Co-Occurring Loss of Function (LOF) Variant: Occurs when a single copy of a gene is lost due to whole gene deletion in the presence of a loss-of-function variant in the same gene.

Homozygous Deletion: Homozygous deletion results when both copies of a gene are lost.

Fusion: Fusion events are gene rearrangements that fuse two previously distinct genes into a single genetic unit. Guardant360 detects fusions in the genes listed in Table 1.

Checkpoint inhibition has been FDA-approved for patients who have no satisfactory alternative treatment option with a tissue TMB score of 10 mut/Mb or higher. In a large clinical study of blood TMB in patients with newly diagnosed non-small cell lung cancer (NSCLC) using Guardant360, Guardant360 TMB score of 16 mut/Mb, 76th percentile, correlates with a tissue TMB score of 10 mut/Mb (Rizvi, N.A., et al. JAMA Oncology, 2020). The distribution of tissue TMB scores has been shown to differ across tumor types (Samstein, R.M, et al. Nature Genetics, 2019). Similarly, in a landscape analysis of Guardant360 TMB, 80th percentile corresponds to the following TMB scores: NSCLC - 20.2 mut/Mb; colorectal cancer - 20.1 mut/Mb; breast cancer- 15.3 mut/Mb; prostate cancer - 13.4 mut/Mb; pancreatic cancer - 11.4 mut/Mb; head and neck squamous cell cancer (HNSCC)- 17.4 mut/Mb; cholangiocarcinoma - 10.5 mut/Mb.

Method and Limitations

Guardant360 sequences 740 cancer-associated genes to identify somatic alterations. Cell-free DNA (cfDNA) is extracted from plasma, partitioned based on methylation status, enriched for targeted regions, and sequenced using the Illumina platform and hg19 as the reference genome. All exons are sequenced in some genes; only clinically significant exons are sequenced in other genes. The types of genomic alterations detected by Guardant360 include single nucleotide variants (SNVs), gene amplifications, fusions/rearrangements, short insertions/deletions (indels, longest detected, 87 base pairs), copy number deletions, promoter methylation and splice site disrupting events (see Table 1). Microsatellite Instability (MSI) is assessed for all cancer types by evaluating somatic changes in the length of repetitive sequences on the Guardant360 panel. A “Not Detected” result in samples where the highest % cfDNA is < 0.2% is an inconclusive result because it does not preclude MSI-High status in tissue. Tumor mutational burden (TMB) score is calculated for all cancer types from somatic SNVs and indels in exons of 497 genes detected in cfDNA, followed by adjusting for tumor shedding levels and the size of the panel. A “Not Evaluable” result is an inconclusive result in samples where the evidence of tumor shedding is insufficient and it does not preclude TMB-High status in tissue. Promoter methylation status is assessed for all cancer types by calculating the % cfDNA methylated molecules overlapping defined promoter regions of 47 genes and is reported as Detected or Not Detected based on the established sensitivity. Promoter methylation may not always be associated with gene inactivation; orthogonal confirmation and clinical correlation are recommended as indicated. Methylation-based tumor fraction is estimated in all cancer types by evaluating multiple differentially methylated regions associated with cancer. The quantifiable range of tumor fractions range from >0.05 to 90%. Tumor fractions less than 0.05% are reported as <0.05%, those higher than 90%, are indicated as >90%. All SNVs and Indels detected on the Guardant360 assay will be evaluated for the likelihood that they are derived from hematopoietic sources by a proprietary algorithm that identifies variants of non-solid-tumor origin that may originate from hematopoietic or non-hematopoietic clonal processes. The test is not intended to be used for the diagnosis of hematolymphoid malignancy. Certain sample or variant characteristics, such as low cfDNA concentration, may result in reduced analytic sensitivity. Guardant360 cannot discern the source of circulating cfDNA, and for some variants in the range of ~40 to 60% cfDNA, the test cannot easily distinguish germline variants from somatic alterations. Guardant360 is not validated for the detection of germline or de novo variants that are associated with hereditary cancer risk. Tissue genotyping should be considered when plasma genotyping is negative, if clinically appropriate.

Table 1: Genes on the Guardant360 Panel

Guardant360 reports single nucleotide variants, splice site mutations, and insertion and deletion variants (indels) in the genes listed below and reports other variant types in select genes as indicated.

ABCB1	BRCC3	CREBBP	ELOC	FGF5	HGF	LGR6	NELFE	PIK3CG	RAF1 [†]	SMARCA4	TOP1
ABL1	BRD2	CRKL	EML4	FGF6	HNF1A	LIG1	NF1 [^]	PIK3R1	RARA	SMARCA1	TOP2A
ABL2	BRD3	CRTC1	EMSY	FGF7	HNRNPDL	LIG4	NF2 [^]	PIK3R2	RASA1	SMARCB1	TOPAZ1
ABRAXAS1	BRD4	CSF1R	EP300	FGF8	HOXB13	LMO1	NFE2L2	PIK3R3	RB1 [^]	SMARCD1	TP53 [^]
ACVR1	BRIP1 [^]	CSF3R	EPCAM	FGF9	HRAS	LRP1B	NFKBIA	PIM1	RBBP6	SMARCE1	TP53BP1
ACVR1B	BSG	CTC1	EPHA3	FGFR1 [†]	HSD3B1	LRP2	NHEJ1	PIN1	RBM10	SMC1A	TP63
ACVR2A	BTG1	CTCF	EPHA5	FGFR2 [†]	HSP90AA1	LRP5	NKX2-1	PKM	RBMY	SMC3	TP73
ADARB2	BTG2	CTLA4	EPHA7	FGFR3 [#]	ICOSLG	LRP6	NOTCH1	PLCG2	RECQL	SMO	TPMT
ADGRA2	BTK	CTNNA1	EPHB1	FGFR4	ID3	LTK	NOTCH2	PLEKHS1	RECQL4	SNCAIP	TRAF2
ADGRG4	BUB1B	CTNNB1	ERBB2 [†]	FH	IDH1	LYN	NOTCH3	PLRG1	RET [#]	SOC1	TRAF3
AFDN	C9orf78	CUL3	ERBB3	FLCN	IDH2	LZTR1	NOTCH4	PMS1	REV3L	SOC3	TRAF7
AGGF1	CALR	CUL4A	ERBB4	FLT1	IDO1	MAD2L2	NOVA1	PMS2 [^]	RGS1	SOS1	TRIM24
AIP	CARD11	CUX1	ERCC1	FLT3	IFNG	MALT1	NPM1	POLA1	RHEB	SOX10	TRIP13
AKT1	CASP8	CWC22	ERCC2	FLT4	IFNGR1	MAP2K1	NPRL2	POLD1	RHOA	SOX17	TSC1
AKT1S1	CASR	CXCR4	ERCC3	FOXA1	IFNGR2	MAP2K2	NPRL3	POLE [^]	RHOB	SOX2	TSC2 [^]
AKT2	CAV1	CYLD	ERCC4	FOXL2	IFNW1	MAP2K4	NRAS	POLH	RICTOR	SOX9	TSHR
AKT3	CBFB	CYP17A1	ERCC5	FOXO1	IGF1	MAP3K1	NRG1 [#]	POLQ	RIF1	SPEN	TSZH2
ALB	CBL	CYP19A1	ERCC6	FOXP1	IGF1R	MAP3K13	NSD1	POT1	RILPL1	SPOP	TYMP
ALK [#]	CBLB	CYP2C19	ERCC6L2	FRS2	IGF2	MAP4K3	NSD2	POU2F2	RIT1	SRC	TYMS
ALOX12B	CCAR1	CYP3A4	ERCC8	FUBP1	IGF2BP3	MAPK1	NSD3	PPARG	RNASEH2B	SRSF2	TYRO3
ALOX15B	CCN6	DAXX	EREG	FUBP3	IGF2R	MAPK3	NSRP1	PIIG	RNF43	SRY	U2AF1
ALOX5	CCNA2	DCUN1D1	ERF	FUS	IKBKE	MAPKAP1	NTHL1	PPM1D	ROBO1	SS18	UBE2T
AMER1	CCNB1	DDIT3	ERG	FYN	IKZF1	MARK2	NTRK1 [#]	PPP2CA	ROBO2	STAG2	UGT1A1
APC [^]	CCND1 [†]	DDR1	ERF1	FZD1	IL1R1	MAX	NTRK2 [#]	PPP2R1A	ROS1 [#]	STAT1	UIMC1
APEX1	CCND2 [†]	DDR2	ESR1 [†]	FZD10	IL2RA	MCL1	NTRK3 [#]	PPP2R2A	RPA1	STAT3	ULBP1
APLN	CCND3	DDX17	ETS1	FZD2	IL2RB	MDC1	NUMA1	PPP3CA	RPS27A	STAT4	ULBP3
AR [†]	CCNE1 [†]	DDX18	ETV1	FZD3	IL2RG	MDM2	NUMB	PPP6C	RPS6KA3	STK11 [^]	USP28
ARAF	CCNE2	DDX27	ETV4	FZD4	IL7R	MDM4	NUP93	PRDM1	RPS6KB1	STK19	USP7
ARFRP1	CD274	DDX3X	ETV5	FZD5	INHBA	MED12	NUTM1	PREX1	RPS6KB2	STK40	USP9X
ARHGAP35	CD276	DDX41	ETV6	FZD6	INPP4B	MEF2B	P2RY8	PREX2	RPTOR	STN1	VEGFA
ARID1A [^]	CD74	DEPDC5	EWSR1	FZD7	INTS6L	MEN1	PABPC1	PRKAR1A	RRAGC	SUFU	VEGFB
ARID1B [^]	CD79A	DEPTOR	EXO1	FZD8	IRF1	MERTK	PAK1	PRKCI	RSP01	SYK	VHL
ARID2	CD79B	DHX15	EZH1	FZD9	IRF2	MET [†]	PAK3	PRKDC	RSP02	SYNCRIP	VIRMA
ASXL1	CDC27	DHX16	EZH2	GAS6	IRF4	MGA	PALB2 [^]	PRKN	RSP04	TACSTD2	WBP11
ATM [^]	CDC5L	DHX36	FAAP100	GATA1	IRS2	MGMT [^]	PARG	PRMT5	RUNX1	TAF1L	WEE1
ATMIN	CDC7	DHX9	FAAP20	GATA2	JAK1	MITF	PARP1	PRPF40B	RUNX1T1	TAP1	WRN
ATR [^]	CDC73	DICER1	FAAP24	GATA3	JAK2	MKNK1	PARP2	PRPF4B	RXRA	TAP2	WT1
ATRX	CDH1 [^]	DIS3L2	FANCA [^]	GATA4	JAK3	MLH1 [^]	PAX3	PSENNEN	RYBP	TAPBP	WWP1
AURKA	CDH6	DLL4	FANCB	GATA6	JUN	MLH3	PAX5	PSMB10	SAMHD1	TBC1D7	XBP1
AURKB	CDK11A	DNAJB1	FANCC	GEN1	KAT6A	MLST8	PAX7	PSMB8	SDC4	TBX3	XPA
AURKC	CDK12 [^]	DNMT1	FANCD2	GID4	KAT6B	MPL	PAX8	PSMB9	SDHA [^]	TCERG1	XPC
AXIN1	CDK4 [†]	DNMT3A	FANCE [^]	GLI1	KDM4A	MRAS	PAXIP1	PTCH1	SDHA2	TCF7L2	XPO1
AXIN2	CDK6 [†]	DNMT3B	FANCF [^]	GNA11	KDM5A	MRE11	PBRM1 [^]	PTDSS1	SDHB [^]	TEK	XRCC1
AXL	CDK7	DOT1L	FANCG [^]	GNA13	KDM5B	MSH2 [^]	PCBP1	PTEN [^]	SDHC [^]	TEN1	XRCC2
B2M [^]	CDK8	DPYD	FANCI	GNAQ	KDM5C	MSH3	PCBP2	PTPN11	SDHD [^]	TENT5C	XRCC3
BABAM1	CDKN1A	DUSP4	FANCL [^]	GNAS	KDM6A	MSH6 [^]	PCDH15	PTPN2	SEM1	TERT [†]	XRCC4
BABAM2	CDKN1B [^]	DYNLL1	FANCM	GPATCH8	KDR	MTAP [^]	PDCD1	PTPRD	SERPINB3	TET1	XRCC5
BAP1	CDKN1C	DYRK2	FAS	GPC3	KEAP1 [^]	MTHFR	PDCD1LG2	PTPRS	SERPINB4	TET2	XRCC6
BARD1	CDKN2A [^]	E2F3	FAT1	GREM1	KIN	MTOR	PDE7A	PTPRT	SESN2	TFE3	YAP1
BCL2	CDKN2B	ECT2L	FBXW7	GRIN2A	KIT [†]	MUTYH [^]	PDGFRA [†]	QKI	SETD2	TFRC	YES1
BCL2L1	CDKN2C	EFTUD2	FCGR2A	GSK3B	KLF4	MYB	PDGFRB	RAB35	SF3B1	TGFBR1	ZC3H13
BCL2L2	CEBPA	EGFR [†]	FCGR3A	GSTM1	KLHL6	MYC [†]	PDK1	RAC1	SF3B3	TGFBR2	ZC3H18
BCL6	CELF4	EIF1AX	FEN1	GSTP1	KLHL9	MYCL	PDPK1	RAD18	SH2D1A	THRAP3	ZC3H4
BCOR	CEP295	EIF4A1	FGF1	H3-4	KMT2A	MYCN	PHF6	RAD21	SHLD1	TIA1	ZMYM3
BCORL1	CFAP20	EIF4A2	FGF10	H3F3A	KMT2B	MYD88	PHLPP1	RAD50 [^]	SHLD2	TIPARP	ZNF217
BCR	CHD4	EIF4A3	FGF12	HACD4	KMT2C	MYO1D	PHLPP2	RAD51 [^]	SLC34A2	TMEM127	ZNF703
BIRC5	CHEK1 [^]	EIF4B	FGF14	HDAC2	KMT2D	NAB2	PHOX2B	RAD51B	SLFN11	TMPRSS2	ZNRF3
BLM	CHEK2 [^]	EIF4E	FGF19	HDAC6	KNSTRN	NBN	PIAS4	RAD51C [^]	SLIT2	TNFAIP3	ZRSR2
BMPR1A	CIC	EIF4E2	FGF2	HELQ	KRAS [†]	NCOR1	PIK3C2B	RAD51D [^]	SMAD2	TNFRSF14	
BRAF [†]	CMTM4	ELAVL1	FGF23	HES1	LATS1	NCR1	PIK3CA [†]	RAD52	SMAD3	TNFRSF1A	
BRCA1 [^]	CMTM6	ELAVL2	FGF3	HEY1	LGR4	NCR3	PIK3CB	RAD54L	SMAD4 [^]	TNK2	
BRCA2 [^]	CNOT3	ELF3	FGF4	HEYL	LGR5	NEGR1	PIK3CD	RAE1E	SMARCA2	TNPO1	

[†] Guardant360 reports alterations in the promoter region of this gene.

[#] Guardant360 reports fusion events involving this gene.

[~] Guardant360 reports rearrangements in this gene.

^{*} Guardant360 reports copy number deletions in this gene.

[†] Guardant360 reports amplifications of this gene.

[^] Guardant360 reports promoter methylation of this gene.

About the Test

The Guardant360 assay was developed and its performance characteristics were determined by Guardant Health, Inc. This test has not been cleared or approved by the U.S. Food and Drug Administration (FDA). The FDA has determined that such clearance or approval is not necessary. This test may be used for clinical purposes and should not be regarded as investigational or for research only. Guardant Health's clinical reference laboratory is certified under the Clinical Laboratory Improvement Amendments of 1988 (CLIA) as qualified to perform high complexity clinical laboratory testing. The laboratory report should be interpreted in the context of other clinical information and laboratory, pathology, and imaging studies by a qualified medical professional prior to initiating or changing a patient's treatment plan. The selection of any, all, or none of the drugs associated with potential clinical benefit (or potential lack of clinical benefit) is entirely at the discretion of the treating medical professional. Drug and trial information are based on the diagnosis written on the submitted test request form; this information is not based on any supplemental information provided by the requesting medical professional, including pathology reports or other molecular studies. Some drugs listed in this report may not be approved or cleared by the FDA for the indicated use. Guardant Health makes no endorsement, expressed or implied, of any product, physician, or procedure contained in this report. This report makes no promises or guarantees that a particular medication will affect (or not affect) the clinical outcome of any patient.

Testing Performed at: Guardant Health

Laboratory Director: Martina Lefterova, MD PhD | CLIA ID: 05D2070300 | CAP #: 8765297 | 505 Penobscot Drive, Redwood City, CA, 94063, USA

Additional information is available

Therapeutic annotations are based on publicly available information. A more detailed Guardant360 report (including Additional Clinical Trials, Detailed Therapy Results, Relevance of Detected Alterations, and References) is available through our online portal at portal.guardanthhealth.com. To set up an account, contact Client Services at [855.698.8887](tel:855.698.8887) or clientservices@guardanthhealth.com.

If you would like to receive this additional information for this report, or with every Guardant360 report, contact Client Services.